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To: Department of Health and Human Services (HHS)
Assistant Secretary for Technology Policy (ASTP)
Office of the National Coordinator for Health Information Technology (ONC)
Attention: Request for Information: HHS Health Sector Artificial Intelligence (AI) RFI

Mary E. Switzer Building, Mail Stop: 7033A
330 C Street SW
Washington, DC 20201

*Submitted electronically to: http://www.regulations.gov*
**Re: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care (HHS Health Sector AI RFI)**

Dear Assistant Secretary Keane,

Please find a response to the HHS Health Sector AI RFI from Anatomy Mapper OpCo below:

## Executive Summary

To accelerate safe and scalable adoption of artificial intelligence (AI) in clinical care, HHS should prioritize interoperable, longitudinal physical health records that link diagnoses, findings, procedures, imaging, and claims to standardized anatomical intelligence.

AI cannot reliably support value-based care, prior authorization reform, price transparency, predictive modeling, or patient-facing transparency when the physical context of care remains embedded in unstructured narrative. Strengthening semantic interoperability within existing frameworks (including USCDI, US Core, and USCDI+) and aligning CMS incentives around structured longitudinal anatomical linkage will improve explainability, enhance productivity, reduce administrative burden, improve claims tracking and reimbursement policy, enable precision medicine, and enable predictive analytics.

Anatomy Mapper OpCo respectfully urges ONC and CMS to treat structured, precise anatomical linkage as foundational AI infrastructure.

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## I. The Infrastructure Gap: Longitudinal Anatomical Context

The principal barriers to scalable, trustworthy clinical AI are not model capability or data quantity. Rather, barriers include poor quality, fragmented, imprecise data lacking standardized, longitudinal physical context.

Diagnoses, procedures, imaging, orders, observations, and claims are frequently recorded as isolated events without a consistent anatomical backbone linking them over time. As a result:

- Episode-based accountability is weakened
- Predictive models lack physical specificity
- Quality benchmarking becomes imprecise
- Documentation must be repeatedly clarified
- AI summarization and analysis lacks structured grounding

AI systems cannot reason effectively about care without knowing precisely where interventions or findings occur and how they relate longitudinally.

Anatomy Mapper models provide a structured anatomical intelligence layer that semantically links clinical and administrative data across encounters. Our validated, semantically interoperable language and our visual documentation models that generate AI-ready data are already in clinical use.  A public demo or preview can be accessed at [https://edu.anatomymapper.com](https://edu.anatomymapper.com).  Anatomy Mapper models are also compatible with current (e.g. ICD-10) and future standards (e.g. ICD-11), and generate codeable concepts that are interoperable across over 100 languages and standards.

By anchoring diagnoses, procedures, and related data to standardized anatomical representations, health records become longitudinally coherent, interoperable, and AI-ready.

## II. CMS Alignment: Longitudinal Accountability & Administrative Simplification

CMS increasingly relies on longitudinal accountability across ACOs, Medicare Advantage, and episode-based payment models. Structured anatomical linkage directly supports these priorities.

When procedures and diagnoses are anchored to standardized anatomical intelligence, CMS and participating organizations can:

- Track procedures at precise anatomical targets over time
- Distinguish recurrence from new pathology
- Evaluate repeat utilization within defined episodes
- Improve longitudinal outcome measurement

Structured anatomical linkage also supports administrative simplification:

- *Claims tracking:* Greater documentation clarity reduces coding ambiguity and rework.
    - **Current state:** Existing procedural coding standards used by CMS, including CPT, were designed primarily for reimbursement abstraction, not longitudinal analytics or AI-enabled care modeling. As a result, many codes aggregate distinct anatomical regions, omit laterality granularity, and compress clinically meaningful variation into single descriptors. While sufficient for billing purposes, this structure is inadequate for modern AI-driven utilization tracking, episode modeling, precision medicine, or predictive analytics. Complementing existing procedural codes with structured anatomical precision enables CMS to preserve reimbursement stability while materially improving analytical clarity.
- *Price transparency:* Procedures defined with anatomical precision improve cross-site comparability and episode grouping.
- *Prior authorization:* High-cost therapies (e.g. biologic medications) and procedures can be evaluated more consistently when clinical indications are linked to structured anatomical context.

Reducing ambiguity across documentation, claims, and authorization workflows materially lowers administrative friction.  This improves productivity for clinicians and payers while strengthening value-based models.

## III. Strengthening Semantic Interoperability (ONC Alignment)

Interoperability must extend beyond transport to semantic precision.

USCDI, US Core, and USCDI+ establish essential data exchange foundations. However, their utility for AI depends on consistent semantic linkage between diagnoses, procedures, findings, observations and anatomical context.

Enhancing semantic normalization through structured anatomical intelligence would:

- Improve longitudinal continuity of physical health records
- Strengthen cross-EHR consistency
- Enable AI-ready normalization of core clinical concepts
- Support responsible post-deployment AI monitoring

Semantic interoperability ensures that both AI systems and humans operate from the same structured representations of care.

## IV. Precision Medicine and Predictive Modeling

AI is increasingly used to predict disease progression and evaluate response to therapy, including biologics and chemotherapeutic regimens.  These capabilities require a trusted and consistent physical data foundation. When longitudinal clinical data are semantically anchored to standardized anatomy, predictive models can:

- Distinguish new disease from recurrence at specific anatomical sites
- Model progression within defined physical regions
- Evaluate therapy response over time
- Support individualized care planning

Without structured anatomical linkage, predictive models remain limited to broad diagnostic categories and fragmented encounter data.

Structured anatomical intelligence serves as a reliable source of truth for AI-enabled precision medicine and predictive analytics.

## V. Administrative Burden Reduction and Productivity

Reducing administrative burden is central to accelerating AI adoption.

Structured anatomical linkage reduces:

- Documentation ambiguity
- Repetitive narrative clarification
- Prior authorization back-and-forth
- Claims reconciliation work
- Manual chart abstraction

It also streamlines quality reporting, utilization review, and risk adjustment workflows by ensuring that diagnoses and procedures are consistently linked to structured physical context. When clinical intent is clearly represented once in a standardized format, downstream systems do not require repeated reinterpretation of narrative documentation. This reduces duplicative data entry, minimizes denials related to incomplete context, and decreases the time clinicians and staff spend reconciling records across encounters.

Anatomy Mapper models also enable efficient, visual clinical workflows that reduce documentation burden and cognitive load. By allowing clinicians to document care through structured, anatomically anchored visual interfaces, repetitive text entry is minimized and contextual understanding improves.

AI systems built on structured data can automate summarization and cross-reference clinical context without requiring clinicians to restate information across notes, claims, and authorizations.

## VI. Patient-Facing Transparency and Understandability

CMS and ONC have emphasized patient access to and understanding of health data. Structured anatomical intelligence enhances both machine interoperability and human comprehension.

When health data are longitudinally linked, semantically normalized, and visualizable:

- Patients can download coherent, interpretable health records
- Conditions and procedures can be visually represented
- Clinical summaries become clearer
- Patients better understand where care occurred and why

Visual, standardized anatomical representations support patient engagement and align with federal priorities around digital health access and transparency.

## VII. Recommended Actions

To accelerate safe AI adoption, HHS should:

- Recognize interoperable, longitudinal anatomical health records as foundational AI infrastructure.
- Encourage ONC guidance that strengthens semantic precision within USCDI, US Core, and USCDI+.

- Align CMS payment models to reward structured documentation quality and measurable administrative burden reduction.
- Promote AI transparency and provenance within interoperable workflows.
- Support benchmark datasets grounded in standardized longitudinal anatomical intelligence.

## Conclusion

Accelerating AI in clinical care requires more than deploying models. It requires interoperable, longitudinal physical health records that connect diagnoses, procedures, and claims to structured anatomical intelligence.

By aligning ONC interoperability policy and CMS payment reform around semantic precision, longitudinal anatomical linkage, administrative simplification, and patient-facing transparency, HHS can materially improve value-based care, reduce burden, enhance productivity, and strengthen public trust in AI-enabled healthcare.

Respectfully submitted,

/Matthew A. Molenda, MD, MBA/

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**References and Links:**

**EDU Version** (Preview or Register to Demo): https://edu.anatomymapper.com

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**Case Studies** enabled by AI-ready data: https://edu.anatomymapper.com/case-studies

See how *Anatomy Mapper standards and models enable interoperability in CMS Aligned Networks*: https://www.youtube.com/watch?v=9WxuM0G4Rlo